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Task Description:

Understanding of qualitative phenomena based on experiments supported by accurate computational simulations (steady and unsteady) for quasi-steady annular/stratified internal condensing flows in micro-gravity and terrestrial environments is essential for good design and performance of condensers in applications such as Rankine-cycle based power (nuclear) plants, Thermal Management systems, Looped Heat Pipes, Capillary Pumped Loops, etc. Condenser-sections in these devices often achieve complete or incomplete condensation by the exit under unspecified or specified (as being proposed) exit pressure conditions while the inlet (pressure and flow rate) and wall conditions are typically always known or specified. These condensing flows' attainability/stability of steady flows, intrinsic waviness, noise-sensitivity (primarily due to minuscule transverse vibrations of the condensing surface), and exit-condition sensitivity critically depend on the presence or absence of gravity in driving the condensate.
Zero-gravity simulations establish that the parameter zone within which steady flows exist for any space-based condenser, experiencing full or partial condensation with unspecified (free) exit condition (as this is a typical exit condition for condensers in many thermal systems), is very small to non-existent. Fully condensing (or boiling) steady flows in zero- or micro-gravity under unspecified exit pressure conditions (for situations where only the heat-transfer surface temperature is known) is actually impossible. This fact is not known as operation of condensers (or boilers) in this mode is often proposed for many space-based thermal system designs. Thus achieving steady flows in such space-based condenser employments is almost impossible unless new technology – as suggested by the current and planned research on exit- pressure condition based flow categorizations - is employed.

The ground based experiments and supporting simulations (both for terrestrial and micro-gravity environments) are to explore how exit-condition effects can lead to different steady and unsteady/oscillatory flows. The unsteady/oscillatory flows may arise from changes in flow regime within the condenser or from system-instabilities (due to interaction of condenser with other upstream or downstream components of the system).

Although the proposed research involves fundamental investigations of flows inside relatively simpler tube and channel geometries, the qualitative understandings/results and tools that will be obtained are going to be sufficiently fundamental and general in nature that they will provide an important framework for facilitating the design, incorporation, and operation of new realistic space-based or ground-based condensers (that is condensers that employ more complex but realistic geometries such as SWEP/Tranter plate heat-exchangers, shell-and-tube type, etc.) in planned systems. In fact the basic result regarding a need for independent pressure-difference imposition (or exit pressure specification if inlet pressure is known) is general enough that it applies to flow-boilers as well.

Task Objectives

Experimental

• Complete experimental investigation of attainability/stability and exit-condition issues for downward and steady internal condensing flows inside a vertical tube. Following this gravity driven condensate flow investigations, initiate similar experimental investigations of gravity-to-shear driven internal condensing flows in inclined-to-horizontal channel (i.e. rectangular cross-section of suitable aspect-ratio) geometries which experience condensation on its bottom wall.

• Ensure that the ground based experimental facility is well instrumented and progressively capable of providing more and more detailed information that can be synthesized with results obtained from the computational capability being developed for these flows.

• Complete ongoing invention efforts and develop a non-intrusive film thickness sensor (based on principles of fluorescence and fiber-optics) that is capable of measuring time-varying thickness of the dynamic condensate. Verify such an invention by independent thickness measurements. Make progress on employing such sensors for the gravity driven annular flow cases in a vertical tube geometry and/or inclined-to-horizontal channel (i.e. rectangular cross-section of suitable aspect-ratio) geometry that experiences condensation on its bottom wall.

Computational

• Complete development of accurate simulation tool for laminar vapor/laminar condensate internal condensing flows for channel or vertical tube geometries.

• Ensure that the simulation tool accurately locates the interface (under smooth or wavy conditions) for terrestrial and zero-gravity environments and meets various convergence and grid independence criteria for both specified and unspecified exit condition cases.

• Ensure that the simulation tool is capable of reliably reproducing known results for classical benchmark problems (such as Nusselt problem of film condensation on a vertical plate, forced vapor flow experiencing film condensation on a horizontal plate, etc.) and, also, capable of providing new non-linear stability results consistent with known experimental results for these classical problems.

• The verifications of the two dimensional code on classical problems as well as through successful comparisons of the simulation results for the ongoing ground based experiments are necessary for further enhancements (compressibility effects under specified exit conditions, three dimensional simulations, etc.) of the code. The verifications will also make simulation tool based zero gravity predictions more trustworthy.

• Eventually the simulation tool should be reliable enough to help plan proper space based experiments and, also, to help reduce the number of design iterations needed to have a condenser (or boiler) operate successfully in a zero gravity environment.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Reliable existing zero-gravity simulations indicate that the parameter zone within which steady flows exist for any space-based condenser that experiences full or partial condensation with an unspecified (free) exit condition (as this is a typical exit condition for condensers in many thermal systems) is often non-existent to small. Thus achieving steady flows in any space-based condenser employments is very difficult (if not next to impossible) unless new technology – as suggested by the current and planned research on specified exit-condition scenarios - is employed.
Experiments: In addition to providing critical and relevant scientific information, the ongoing/proposed experiments will also establish that condensers (and boilers) in space must be operated under specified pressure-difference conditions. This critical understanding will facilitate development of space-based two-phase thermal system technologies (air-conditioning, refrigeration, Rankine Power cycle, etc.) as well as novel ground based micro-scale cooling technologies.

Software: The ongoing/proposed simulation tool component will have a far reaching impact as it will shed light on the physics of an array of related problems (viz. boiling, condensing, and evaporating phase-change flows of pure fluids) – both in simple geometries of fundamental interest and complex geometries of engineering interest. Furthermore, eventual inclusion of diffusion equations (available through COMSOL's CFD tools) in the simulation tool can allow one to model diffusion processes in vapor and/or liquid phases. Examples of innovative research in the area of phase-change flows involving either of the two types of diffusion are: condensation in the presence of non-condensable gas and evaporation of saline liquid films. Both these simulation abilities will assist in improving technologies related to cleaning/recycling of air and water for long term human presence in reduced and micro-gravity environments.

Earth Benefits

The exit condition based flow categorizations and experimental results are also going to be useful for many new terrestrial systems (such as two-phase electronic or chip cooling) that need to employ a condenser with a steady and stable flow for the thermal system. While it is true that many terrestrial systems can be made to function by sufficient number of trial and error experiments or by an on-off strategy that does not require attainment of steady flows, the ground based results from this research will help significantly in innovation and improvement of superior new technologies for such systems. For example, a specified exit condition mode of operation (based on electronic flow control) is expected to allow steady and stable operation of many terrestrial condensers in novel systems (electronic cooling, etc.) with the ability to handle a larger range of heat-load variations and cutting down on the system's start up times.

Also, the proposed (nearly complete) development of film thickness sensor will be valuable in the measurement processes that are important to the development of many terrestrial applications that involve film cooling, film evaporation, film flows in chemical process industries, etc.

Task Progress:

Earlier reported experimental and computational results for quasi-steady partially condensing flows confirmed that both the flow features and heat transfer rates inside a condenser depended on the specification of inlet (pressure, temperature, and velocity), wall-temperature, and exit pressure conditions. Recent experimental results confirm the same for fully condensing quasi-steady flows .
The fundamental/scientific result – supported by our experiments and computations/theory – states that forced steady internal flow condensation (or flow boiling) for known inlet conditions (pressure, velocity, and temperature), under the more common (and useful) case of specified or known temperature conditions for the heat-transfer surface, can always be realized under a suitably specified pressure-difference across the device. We further show that this pressure-difference specification is feasible but not critical for gravity driven condensate cases. However, our computational results show that operation of shear driven condensing flows in 0g (or micro-g or normal transverse g) under specified pressure-difference conditions is both feasible and critically needed for desired realizations of shear/pressure driven flows in shear driven condensers (or boilers). The classical Koh and Nusselt problems' steady solutions, when interpreted as a case involving "zero" specified pressure difference, fit the above result whenever a steady film or annular/stratified steady condensing flow is feasible.

A far reaching implication of the above is that, unlike single-phase and air-water flows, phase-change flow realizations under specified heat-flux conditions (often investigated because they do not require careful exit pressure specifications) are quite different from realizations under specified temperature conditions (these cases, to our knowledge, have not been investigated despite their greater usefulness).

For fully condensing flows, we are successful in developing appropriate electronic flow control strategies and experimentally establishing the above stated result that is vital to the design of many novel closed-flow-loop thermal systems that involve flow boilers and flow condensers experiencing a complete change in phase. The result implies: (i) the heat-flux and temperature boundary conditions are known to be equivalent to one another for single-phase or air-water flows but these two types of boundary conditions are not equivalent to one another for flow condensers (and boilers), particularly if a complete change in phase occurs; (ii) for boilers and condensers, under heat-flux specification on the phase-change surface, one can specify only inlet (temperature, pressure, and velocity) and wall conditions and expect a well defined steady flow realization/solution (as is the case for some repeatable experiments in the literature and this is because, under these conditions, steady equations governing the flow are essentially parabolic in the flow direction); and (iii) for boilers and condensers , under temperature specification for the phase-change surface, one must concurrently specify inlet (temperature, pressure, and velocity) conditions and exit pressure to expect a well defined steady flow realization/solution (this is because, under these conditions, steady equations governing the flow are essentially elliptic).

The above established lack of equivalence between the heat-flux and temperature boundary conditions for flow boilers/condensers arises from the extra degree of freedom associated with the placement of the interface. This degree of freedom is mechanically controlled/eliminated, for temperature boundary conditions, by specification of pressure-difference across the device. Equivalently (though with some difficulty in practice), this degree of freedom can also be thermally controlled/eliminated by a specialized method of heating/cooling that prescribes heat-flux conditions (which also leads to specification of interfacial mass-flux for the interface near the heat-transfer surface). For temperature boundary condition, when exit pressure is changed from one steady value to another, the changes required of the interior vapor flow towards achieving a new steady duct flow are such that they do not demand removal of the new exit pressure imposition back to the original steady value. This behavior of condensing flow is counter-intuitive and unlike that of any incompressible single phase duct flow which cannot allow imposition of an exit pressure other than

the one associated with the “required” pressure-difference. Instead, new steady flows may be achieved for different steady imposed pressure-differences or exit pressure prescriptions (termed category I flows) through appropriate changes in the vapor/liquid interface configurations and associated changes in interfacial mass, heat transfer rates (both local and overall), and other flow variables. This special feature of these flows is for the commonly occurring large heat sink situations, as is the case with our experiments, for which the condensing surface temperature (though not heat-flux) remains approximately the same for any given set of inlet conditions while exit condition changes. In this report's experimental context of flows of a pure vapor that experiences film condensation on the inside walls of a vertical tube, the reported results provide important quantitative and qualitative understanding.

The theoretical and experimental results presented in this report allow us to propose important exit-condition based categorization of these flows. Of these, category II flows, which are defined to be the cases for which exit pressures are left unspecified, have typically been assumed to hold for most actual steady realizations of gravity driven as well as zero-gravity flows. However it is shown here that steady flows under self selected exit pressure conditions are achieved under these fully condensing flow conditions for gravity driven flows alone and, even then, only if special hardware arrangements are made that allow vapor to freely choose an exit pressure. If this is not so, one often has an inadvertent category I flow without the explicit knowledge of the imposed pressure-difference. It is theoretically shown here that though this gravity driven category II steady flow situation in the absence of specification of exit condition constitutes an “ill posed” steady boundary value problem, under special arrangements, steady flows may still be experimentally realized. This fact is also validated by unsteady simulation results which show that solutions from many different initial conditions may tend to a unique steady attracting solution (as time tends to infinity) for these internal and/or external flows. These “attractors” exist and are strong for gravity driven flows and are often weak to non-existent for zero gravity or horizontal shear/pressure driven flows (both for internal and external flows) - unless pressure-difference is specified. In micro-gravity, the “attractors” for category II internal flows are weak in the sense that they significantly lose their strength of “attraction” after a very short downstream distance – beyond which the flows are indeterminate if exit pressure remains unspecified. For these micro-gravity situations, the remedy is to run condensers under suitably specified inlet and exit pressures (category I) conditions as well as a proper cooling strategy (i.e. proper wall temperature variations). The details of our hardware and electronic flow-control strategy recommendations on how to run space based condensers under category I flow conditions is described in this report. However additional experimental/theoretical details are part of ongoing research and future reports.

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